

Physics 827: Homework Set No. 2

Due date: Thursday, October 6, 2011, 3:00pm
in PRB M2043 (Biao Huang's office)

Total point value of set: 60 points

Problem 1 (5 pts.): Exercise 1.8.3 (Shankar p.41)

Problem 2 (5 pts.): Exercise 1.8.6 (Shankar p.42)

Problem 3 (5 pts.): Exercise 1.8.8 (Shankar p.43) [Note: in part (2) it should read $M^i M^j = -M^j M^i$]

Problem 4 (5 pts.): Exercise 1.9.2 (Shankar p.55)

Problem 5 (5 pts.): Exercise 1.9.3 (Shankar p.55)

Problem 6 (5 pts.): Exercise 1.10.2 (Shankar p.63)

Problem 7 (5 pts.): Exercise 1.10.3 (Shankar p.63)

Problem 8 (15 pts.): For $\hat{H}$ Hermitean, $\hat{U}, \hat{V}$ unitary, show:

(i) $\hat{U}\hat{H}\hat{U}^{-1}$ is Hermitean

(ii) $(\hat{U})^n$ is unitary

(iii) $\hat{U}\hat{V}$ is unitary

(iv) If $f(x)$ is real and has no poles, $f(\hat{H})$ is Hermitean

(v) $i \frac{\hat{U} - \hat{I}}{\hat{U} + \hat{I}}$ is Hermitean

(vi) $\frac{\hat{I} - i\hat{H}}{\hat{I} + i\hat{H}}$ is unitary

Problem 9 (10 pts.): Show that $e^{\hat{L}}\hat{M}e^{-\hat{L}} = \sum_{n=0}^{\infty} \frac{1}{n!} [\hat{L}, \hat{M}]_{(n)}$ where $[\hat{L}, \hat{M}]_{(0)} = \hat{M}$, $[\hat{L}, \hat{M}]_{(n)} = [\hat{L}, [\hat{L}, \hat{M}]_{(n-1)}]$.

(We will need this later for $\hat{L} = i\hat{H}t$ where $\hat{H}$ is the Hamiltonian.)

Hint: Consider $\hat{F}(\alpha) = e^{\alpha\hat{L}}\hat{M}e^{-\alpha\hat{L}}$ and expand it into a Taylor series around the origin.

Exercise 1.8.3 (p. 41)

(1) Characteristic polynomial: $\det(\hat{\Omega} - \omega \hat{I}) = 0$

$$\begin{vmatrix} 1-\omega & 0 & 0 \\ 0 & \frac{3}{2}-\omega & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2}-\omega \end{vmatrix} = (1-\omega) \left[\left(\frac{3}{2}-\omega \right)^2 - \frac{1}{4} \right] = (1-\omega) (2-3\omega+\omega^2)$$

$$= (1-\omega)(2-\omega)(1-\omega) = 0 \Rightarrow \boxed{\omega_1 = \omega_2 = 1, \omega_3 = 2.} \checkmark$$

(2) Eigenvectors: represent $|\omega=2\rangle$ as column vector $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = 2 \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \Rightarrow \begin{cases} a_1 = 2a_1 \Rightarrow a_1 = 0 \\ 3a_2 - a_3 = 4a_2 \Rightarrow -a_3 = a_2 \\ -a_2 + 3a_3 = 4a_3 \Rightarrow -a_2 = a_3 \end{cases} \text{ redundant.}$$

$\Rightarrow |\omega=2\rangle \leftrightarrow \begin{pmatrix} 0 \\ a \\ -a \end{pmatrix} (a \in \mathbb{R})$; The normalized eigenvector

is $|\omega=2\rangle \leftrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \checkmark$ but any vector of the form $\begin{pmatrix} 0 \\ a \\ -a \end{pmatrix}$ is

an eigenvector, too, with the same eigenvalue.

(This differs from the general form given in the problem statement which is, in fact, not the general solution, but the normalized one.)

(3) $|\omega=1\rangle$ eigenvectors:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \Rightarrow \begin{cases} a_1 = a_1 \\ 3a_2 - a_3 = 2a_2 \Leftrightarrow a_2 = a_3 \\ -a_2 + 3a_3 = 2a_3 \Leftrightarrow a_3 = a_2 \end{cases} \text{ redundant}$$

$\Rightarrow |\omega=1\rangle \leftrightarrow \begin{pmatrix} a_1 \\ a_2 \\ a_2 \end{pmatrix}$ general eigenvector;

normalized eigenvector $|\omega=1\rangle \leftrightarrow \begin{pmatrix} b \\ c \\ c \end{pmatrix} \cdot \frac{1}{\sqrt{b^2+2c^2}}$

($a, b \in \mathbb{R}$ arbitrary) $\rightarrow$ This is a 2-d subspace.

(4) Find two orthonormal eigenvectors in the subspace corresponding to $\omega=1$:

Can start by choosing $|\omega=1,1\rangle \leftrightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ ($b=1, c=0$)

Orthonormalize $|\omega=1,2\rangle$ on this state with Gram-Schmidt procedure:

$$|\omega=1,2\rangle = \begin{pmatrix} b \\ c \\ c \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} (1,0,0) \begin{pmatrix} b \\ c \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ c \\ c \end{pmatrix}$$

Normalized solution: $|\omega=1,2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

(Many other solutions are possible)

Exercise 1.8.6 (p.42)

- (1) Show that for Hermitian $\hat{\Omega}$, $\det \hat{\Omega} = \prod_{i=1}^n \omega_i$ where $(\omega_i, i=1, \dots, n)$ are the eigenvalues of $\hat{\Omega}$

Answer: Since $\hat{\Omega}$ is Hermitian, a unitary matrix $\hat{U}$ can be constructed such that $\hat{\Omega}' = \hat{U} \hat{\Omega} \hat{U}^{-1} = \begin{pmatrix} \omega_1 & & \\ & \omega_2 & \\ & & \ddots \\ & & & \omega_n \end{pmatrix}$ is diagonal, with eigenvalues on the diagonal.

$$\begin{aligned} \Rightarrow \det \hat{\Omega} &= \det \hat{\Omega} \cdot \det(\hat{U}^{-1} \hat{U}) = \det(\hat{\Omega} \hat{U}^{-1} \hat{U}) \\ &= \det(\hat{U} \hat{\Omega} \hat{U}^{-1}) = \det(\hat{\Omega}') = \prod_{i=1}^n \omega_i \quad \checkmark \end{aligned}$$

$$(2) \quad \text{tr} \hat{\Omega} = \text{tr}(\hat{\Omega} \hat{U}^{-1} \hat{U}) = \text{tr}(\hat{U} \hat{\Omega} \hat{U}^{-1}) = \text{tr}(\hat{\Omega}') = \sum_{i=1}^n \omega_i$$

$$\text{tr}(\hat{\Omega} \hat{U}^{-1} \hat{U}) = \sum_{i,j,k} \Omega_{ij} (\hat{U}^{-1})_{jk} U_{ki} = \sum_{k,j,i} U_{ki} \Omega_{ij} (\hat{U}^{-1})_{jk} = \text{tr}(\hat{U} \hat{\Omega} \hat{U}^{-1})$$

Exercise 1.8.8 (p. 43)

$$\hat{M}^i \hat{M}^j + \hat{M}^j \hat{M}^i = 2\delta^{ij} \hat{I} \quad (i, j = 1, \dots, 4), \quad \hat{M}^i \text{ Hermitian}$$

(1) For $i=j$, this gives

$$2(\hat{M}^i)^2 = 2\hat{I} \Rightarrow (\hat{M}^i)^2 = \hat{I} \quad (i=1, \dots, 4)$$

$\Rightarrow$ each $\hat{M}^i$ is its own inverse!

In the eigenbasis of $\hat{M}^i$ (a different basis for each i !)

this becomes

$$\begin{pmatrix} (m_1^i)^2 \\ (m_2^i)^2 \\ \vdots \\ (m_n^i)^2 \end{pmatrix} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & \ddots \\ & & & 1 \end{pmatrix} \Rightarrow (m_j^i)^2 = 1, \begin{pmatrix} i=1, \dots, 4 \\ j=1, \dots, n \end{pmatrix}$$

Since the $\hat{M}^i$ are Hermitian, all eigenvalues m_j^i are real $\Rightarrow$ $\boxed{m_j^i = \pm 1 \text{ for } i=1, \dots, 4; j=1, \dots, n}$ ✓

(2) For $i \neq j$ we have $\hat{M}^i \hat{M}^j + \hat{M}^j \hat{M}^i = 0$

Multiply this with $\hat{M}^i$ or $\hat{M}^j$, from left or right;
for example,

$$(\hat{M}^i)^2 \hat{M}^j + \hat{M}^i \hat{M}^j \hat{M}^i = 0$$

Take trace and use $(\hat{M}^i)^2 = \hat{I}$ from (1) above:

$$\text{tr}((\hat{M}^i)^2 \hat{M}^j) + \text{tr}(\hat{M}^i \hat{M}^j \hat{M}^i) = \text{tr}(\hat{M}^j) + \text{tr}((\hat{M}^i)^2 \hat{M}^j)$$

$$= 2 + \text{tr}(\hat{M}^j) = 0, \quad j=1, \dots, 4$$

$\xrightarrow{\text{proved in 1.8.6(2)}}$

$$\Rightarrow \boxed{\text{tr}(\hat{M}^i) = 0, \quad i=1, \dots, 4} \quad \checkmark$$

(3) From 1.8.6 (2) we know that

$$0 = \text{tr}(\hat{M}^i) = \sum_{j=1}^n m_j^i = \sum_{j=1}^n (\pm 1)$$

This can only be achieved if the sum has an equal number of +1 and -1 eigenvalues, which requires an even number of terms $\Rightarrow \boxed{n = \text{even}}$

The matrices must be even-dimensional

Exercise 1.9.2 (p.55)

$$\text{If } \hat{H} = \hat{H}^\dagger, \quad \hat{U}^\dagger = (e^{i\hat{H}})^\dagger = \sum_{n=0}^{\infty} \frac{[(i\hat{H})^n]^\dagger}{n!} = \sum_{n=0}^{\infty} \frac{(-i\hat{H}^\dagger)^n}{n!} \stackrel{(\hat{H}=\hat{H}^\dagger)}{=} \sum_{n=0}^{\infty} \frac{(-i\hat{H})^n}{n!} \\ = e^{-i\hat{H}} = \hat{U}^{-1} \quad \text{since } \hat{U}^\dagger \hat{U} = e^{i\hat{H}} e^{-i\hat{H}} = e^{i\hat{H}-i\hat{H}} = e^0 = \hat{I} \\ \text{since } [\hat{H}, \hat{H}] = 0$$

$$\Rightarrow \boxed{\hat{U} = e^{i\hat{H}} \text{ is unitary if } \hat{H} \text{ is Hermitian}} \quad \checkmark$$

Exercise 1.9.3 (p.55)

Since $\hat{H}$ is Hermitian, it can be diagonalized by a unitary transformation to its eigenvalue basis:

$$\hat{H}' = \hat{V} \hat{H} \hat{V}^{-1} = \begin{pmatrix} E_1 & & \\ & E_2 & \\ & & \ddots \\ & & & E_n \end{pmatrix} \quad (\hat{V} \text{ unitary})$$

Under this transformation

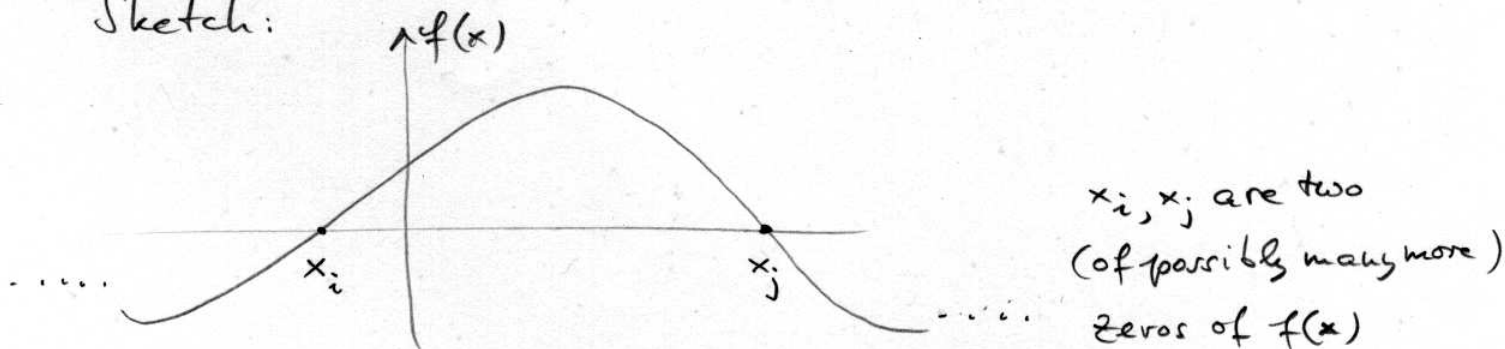
$$\hat{U} \mapsto \hat{U}' = e^{i\hat{H}'} = e^{i\hat{V}\hat{H}\hat{V}^{-1}} = \sum_{n=0}^{\infty} \frac{(i\hat{V}\hat{H}\hat{V}^{-1})^n}{n!} \\ = \sum_{n=0}^{\infty} \frac{\overbrace{i\hat{V}\hat{H}\hat{V}^{-1} i\hat{V}\hat{H}\hat{V}^{-1} \dots i\hat{V}\hat{H}\hat{V}^{-1}}^{n \text{ times}}}{n!} \\ = \sum_{n=0}^{\infty} \frac{\hat{V} \underbrace{(i\hat{H} i\hat{H} \dots i\hat{H})}_{n \text{ times}} \hat{V}^{-1}}{n!} \quad (\text{using } \hat{V}^{-1}\hat{V} = \hat{I}) \\ = \hat{V} \left(\sum_{n=0}^{\infty} \frac{(i\hat{H})^n}{n!} \right) \hat{V}^{-1} = \hat{V} e^{i\hat{H}} \hat{V}^{-1} = \hat{V} \hat{U} \hat{V}^{-1}$$

$$\text{Hence } \underline{\det \hat{U}} = \det(\hat{V}^{-1} \hat{U}' \hat{V}) = \det(\hat{V} \hat{V}^{-1} \hat{U}') = \det \hat{U}' \\ = \det(e^{i\hat{H}'}) = \det \begin{pmatrix} e^{iE_1} & & \\ & e^{iE_2} & \\ & & \ddots \\ & & & e^{iE_n} \end{pmatrix} = e^{i \sum_{j=1}^n E_j} = e^{i \text{tr} \hat{H}'} = e^{i \text{tr}(\hat{V} \hat{H} \hat{V}^{-1})} \\ = e^{i \text{tr} \hat{H}} = \underline{\underline{\det \hat{U}}}$$

Exercise 1.10.2 (p.63)

Show $\delta(f(x)) = \sum_i \frac{\delta(x-x_i)}{\left| \frac{df}{dx} \right|_{x=x_i}}$ where x_i are the zeros of $f(x)$.

Sketch:



Near the zeros, $f(x)$ can be approximated by a linear function: $f(x) \approx \left. \frac{df}{dx} \right|_{x_i} (x-x_i)$ (*)

$\left. \frac{df}{dx} \right|_{x=x_i}$ is a (positive or negative) real number.

At every place $f(x)=0$, $\delta(f(x))$ has a singularity. Everywhere else it is zero. So near each zero we can use the linear approximation (*); it becomes inaccurate as we move away from the zero x_i , but that doesn't matter because $\delta(f(x))$ is there zero, irrespective of whether we use the exact form for $f(x)$ or the linear approximation (*). So we can write

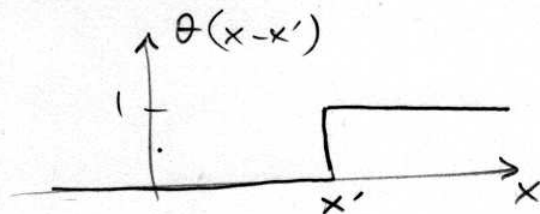
$$\delta(f(x)) = \sum_{\substack{i \\ f(x_i)=0}} \delta\left(\left. \frac{df}{dx} \right|_{x_i} (x-x_i)\right)$$

Using $\delta(ax) = \frac{1}{|a|} \delta(x)$ we arrive at the desired result:

$$\left[\delta(f(x)) = \sum_{\substack{i \\ f(x_i)=0}} \frac{\delta(x-x_i)}{\left| \frac{df}{dx} \right|_{x_i}} \right] \checkmark$$

Exercise 1.10.3 (p. 63)

Show $\delta(x-x') = \frac{d}{dx} \theta(x-x')$



Obviously $\frac{d}{dx} \theta(x-x') = 0$ for $x > x'$ and $x < x'$, i.e. for $x \neq x'$

Also obviously $\theta(x-x')$ has infinite slope at $x=x'$

$\Rightarrow \frac{d}{dx} \theta(x-x') = A \delta(x-x')$ with unknown normalization.

Check

$$\int_{-\infty}^{\infty} dx \frac{d}{dx} \theta(x-x') = \theta(x-x') \Big|_{x=-\infty}^{x=\infty} = \theta(\infty) - \theta(-\infty) = 1 \quad (\text{indep. of } x')$$

$$= \int_{-\infty}^{\infty} dx A \delta(x-x') = A \quad \Rightarrow A = 1 \checkmark$$

Problem 8 ($\hat{H}$ Hermitian ($\hat{H}^\dagger = \hat{H}$); $\hat{U}, \hat{V}$ unitary ($\hat{U}^\dagger = \hat{U}^{-1}, \hat{V}^\dagger = \hat{V}^{-1}$))

(i) $(\hat{U} \hat{H} \hat{U}^{-1})^\dagger = (\hat{U}^{-1})^\dagger \hat{H}^\dagger \hat{U}^\dagger = \hat{U} \hat{H} \hat{U}^{-1}$ Hermitian $\diamond$

(ii) $(\hat{U}^n)^\dagger \hat{U}^n = (\hat{U} \cdot \hat{U} \dots \hat{U})^\dagger (\hat{U} \cdot \hat{U} \dots \hat{U})$
 $= (\hat{U}^\dagger \hat{U}^\dagger \dots \hat{U}^\dagger) (\hat{U} \hat{U} \dots \hat{U}) = \hat{I}$

Similarly $\hat{U}^n (\hat{U}^n)^\dagger = \hat{I} \Rightarrow \hat{U}^n$ is unitary $\diamond$

(iii) $(\hat{U} \hat{V})^\dagger = \hat{V}^\dagger \hat{U}^\dagger = \hat{V}^{-1} \hat{U}^{-1} = (\hat{U} \hat{V})^{-1}$ unitary $\diamond$

(iv) Since $f(x)$ has no singularities, it can be Taylor expanded around any point $x_0 \in \mathbb{R}$ with a finite (non-zero) convergence radius:

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x_0} (x - x_0)^n$$

$$\Rightarrow f(\hat{H}) = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x_0} (\hat{H} - x_0 \hat{I})^n$$

$$\begin{aligned} \Rightarrow [f(\hat{H})]^\dagger &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\left. \frac{d^n f}{dx^n} \right|_{x_0} \right)^* [(\hat{H} - x_0 \hat{I})^n]^\dagger = \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x_0} (\hat{H}^\dagger - x_0 \hat{I})^n = f(\hat{H}^\dagger) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n f}{dx^n} \right|_{x_0} (\hat{H} - x_0 \hat{I})^n = f(\hat{H}) \end{aligned}$$

$\uparrow$ f is real

So if $\hat{H} = \hat{H}^\dagger$, $\underline{f(\hat{H}) = f(\hat{H}^\dagger) = [f(\hat{H})]^\dagger}$ $\diamond$

The proof works for any x_0 , so for any eigenstate of $\hat{H}$ I can expand around the corresponding eigenvalue, proving Hermiticity on that eigenstate. If $\hat{H}$ is Hermitian on all eigenstates, it is Hermitian in the entire Hilbert space.

$$(v) \quad i \frac{\hat{U} - \hat{I}}{\hat{U} + \hat{I}} = i (\hat{U} - \hat{I})(\hat{U} + \hat{I})^{-1} = i (\hat{U} + \hat{I})^{-1} (\hat{U} - \hat{I})$$

Since $(\hat{U} + \hat{I})^{-1}$ commutes with $\hat{U} - \hat{I}$:

$$\begin{aligned} (\hat{U} + \hat{I})(\hat{U} - \hat{I})(\hat{U} + \hat{I})^{-1} &= (\hat{U} + \hat{I})\hat{U}(\hat{U} + \hat{I})^{-1} - (\hat{U} + \hat{I})\hat{I}(\hat{U} + \hat{I})^{-1} \\ &= \hat{U}(\hat{U} + \hat{I})(\hat{U} + \hat{I})^{-1} - \hat{I} = \hat{U} - \hat{I} \end{aligned}$$

$$\text{Hence } \left(i \frac{\hat{U} - \hat{I}}{\hat{U} + \hat{I}} \right)^+ = \left(i (\hat{U} - \hat{I})(\hat{U} + \hat{I})^{-1} \right)^+ = -i [(\hat{U} + \hat{I})^{-1}]^+ (\hat{U} - \hat{I})^+$$

Now $(\hat{U} + \hat{I})^{-1}$ exists as long as none of $\hat{U}$'s eigenvalues equals -1 . Then we can use (iv) and write

$$[(\hat{U} + \hat{I})^{-1}]^+ = [f(\hat{U})]^+ = f(\hat{U}^+) = f(\hat{U}^{-1}) = (\hat{U}^{-1} - \hat{I})^{-1}$$

$$\begin{aligned} \Rightarrow \left(i \frac{\hat{U} - \hat{I}}{\hat{U} + \hat{I}} \right)^+ &= -i (\hat{U}^{-1} - \hat{I})^{-1} (\hat{U} - \hat{I}) = -i (\hat{U}^{-1} + \hat{I})^{-1} (\hat{I} - \hat{U}) \hat{U}^{-1} \\ &= i (\hat{U}^{-1} + \hat{I})^{-1} (\hat{U} - \hat{I}) \hat{U}^{-1} = i (\hat{U} - \hat{I}) (\hat{U}^{-1} + \hat{I})^{-1} \hat{U}^{-1} \\ &= i (\hat{U} - \hat{I}) (\hat{U} (\hat{U}^{-1} + \hat{I}))^{-1} = i (\hat{U} - \hat{I}) (\hat{I} + \hat{U})^{-1} \\ &= i \frac{\hat{U} - \hat{I}}{\hat{U} + \hat{I}} \quad \text{Hermitian} \end{aligned}$$

(vi) The inverse of $\hat{I} + i\hat{H}$ exists since all eigenvalues of $\hat{H}$ are real. Like in (v) you show that $(\hat{I} + i\hat{H})^{-1}$ commutes with $\hat{I} - i\hat{H}$.

$$\begin{aligned} \left(\frac{\hat{I} - i\hat{H}}{\hat{I} + i\hat{H}} \right)^+ &= [(\hat{I} - i\hat{H})(\hat{I} + i\hat{H})^{-1}]^+ = \underbrace{[(\hat{I} + i\hat{H})^{-1}]^+}_{f(\hat{H})} (\hat{I} - i\hat{H})^+ \\ &= (\hat{I} - i\hat{H})^{-1} (\hat{I} + i\hat{H}) = [(\hat{I} + i\hat{H})^{-1} (\hat{I} - i\hat{H})]^{-1} \\ &\quad f((i\hat{H})^+) = f(-i\hat{H}) \quad = \left(\frac{\hat{I} - i\hat{H}}{\hat{I} + i\hat{H}} \right)^{-1} \quad \text{unitary} \end{aligned}$$

Problem 9

Prove $e^{\hat{L}} \hat{M} e^{-\hat{L}} = \sum_{n=0}^{\infty} \frac{1}{n!} [\hat{L}, \hat{M}]_{(n)}$ where $[\hat{L}, \hat{M}]_{(0)} = \hat{M}$,
 $[\hat{L}, \hat{M}]_{(n)} = [\hat{L}, [\hat{L}, \hat{M}]_{(n-1)}]$.

Answer: Consider $\hat{F}(\alpha) = e^{\alpha \hat{L}} \hat{M} e^{-\alpha \hat{L}}$ ($\alpha \in \mathbb{R}$)

Expand $\hat{F}(\alpha)$ into a Taylor series around $\alpha = 0$:

$$\hat{F}(\alpha) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{d^n \hat{F}}{d\alpha^n} \right)_{\alpha=0} \alpha^n \quad (\hat{F}(0) = \hat{M} = [\hat{L}, \hat{M}]_{(0)})$$

Now $\frac{d}{d\alpha} (e^{\alpha \hat{L}}) = e^{\alpha \hat{L}} \hat{L}$ and $\frac{d}{d\alpha} e^{-\alpha \hat{L}} = -\hat{L} e^{-\alpha \hat{L}}$

Hence

$$\begin{aligned} \frac{d}{d\alpha} \hat{F}(\alpha) &= \frac{d}{d\alpha} (e^{\alpha \hat{L}} \hat{M} e^{-\alpha \hat{L}}) = (e^{\alpha \hat{L}} \hat{L}) \hat{M} e^{-\alpha \hat{L}} + e^{\alpha \hat{L}} \hat{M} (-\hat{L} e^{-\alpha \hat{L}}) \\ &= e^{\alpha \hat{L}} [\hat{L}, \hat{M}] e^{-\alpha \hat{L}} \xrightarrow{\alpha=0} [\hat{L}, \hat{M}] = [\hat{L}, [\hat{L}, \hat{M}]_{(0)}] = [\hat{L}, \hat{M}]_{(1)} \end{aligned}$$

$$\begin{aligned} \frac{d^2}{d\alpha^2} \hat{F}(\alpha) &= \frac{d}{d\alpha} (e^{\alpha \hat{L}} [\hat{L}, \hat{M}]_{(1)} e^{-\alpha \hat{L}}) = (e^{\alpha \hat{L}} \hat{L}) [\hat{L}, \hat{M}]_{(1)} e^{-\alpha \hat{L}} \\ &\quad - e^{\alpha \hat{L}} [\hat{L}, \hat{M}]_{(1)} \hat{L} e^{-\alpha \hat{L}} \\ &= e^{\alpha \hat{L}} [\hat{L}, [\hat{L}, \hat{M}]_{(1)}] e^{-\alpha \hat{L}} = e^{\alpha \hat{L}} [\hat{L}, \hat{M}]_{(2)} e^{-\alpha \hat{L}} \xrightarrow{\alpha=0} [\hat{L}, \hat{M}]_{(2)} \end{aligned}$$

etc. for higher derivatives

$$\Rightarrow \left. \frac{d^n \hat{F}}{d\alpha^n} \right|_{\alpha=0} = [\hat{L}, \hat{M}]_{(n)} \Rightarrow \hat{F}(\alpha) = \sum_{n=0}^{\infty} \frac{1}{n!} [\hat{L}, \hat{M}]_{(n)} \alpha^n$$

$$\Rightarrow \boxed{e^{\hat{L}} \hat{M} e^{-\hat{L}} = \hat{F}(1) = \sum_{n=0}^{\infty} \frac{1}{n!} [\hat{L}, \hat{M}]_{(n)}} \quad \diamond$$